

transfer learning and supervised learning are employed to enhance the model's performance.

Model integration: The final step is to integrate the trained SLM with existing systems. This includes deploying the model on cloud platforms, setting up APIs for access, and ensuring seamless integration with other applications.

Microsoft Azure's native services for building an SLM

Microsoft Azure provides a comprehensive suite of services that can be utilised to build and deploy an SLM.

Azure Blob Storage: Azure Blob Storage is used to store the raw and processed data. It provides scalable and secure storage for large datasets.

Azure Databricks: Azure Databricks is an analytics platform that facilitates data preprocessing, cleaning, and transformation. It integrates seamlessly with other Azure services and supports various data processing frameworks.

Azure Machine Learning: Azure Machine Learning is used to train and fine-tune the SLM. It provides a range of tools and libraries for model training, validation, and deployment.

Azure Cognitive Services: Azure Cognitive Services offers pre-trained LLMs that can be leveraged for data filtering and classification. These services include text analytics, language understanding (LUIS), and custom vision.

Azure API Management: Azure API Management is used to manage and deploy APIs for accessing the SLM. It provides security, scalability, and monitoring capabilities.

Real-time use cases in healthcare

The healthcare sector stands to benefit significantly from the implementation of SLMs. Some real-time use cases include:

Medical diagnosis: An SLM can analyse patient records, lab results, and medical literature to assist doctors in diagnosing complex conditions. For instance, it can help identify rare diseases by comparing symptoms with documented cases and suggesting possible diagnoses that may not be immediately obvious.

Drug discovery: Pharmaceutical companies can utilise SLMs to sift through vast amounts of scientific research, patents, and clinical trial data to identify potential drug candidates. By analysing patterns and correlations, the model can suggest novel compounds or repurpose existing drugs for new therapeutic uses.

Patient engagement: An SLM can enhance patient engagement by providing personalised health information and recommendations. For example, a virtual health assistant powered by the SLM can generate tailored advice based on a patient's medical history, lifestyle, and preferences, improving adherence to treatment plans.

Clinical research: Researchers can use SLMs to extract relevant information from a multitude of research papers and clinical trial reports. This can facilitate meta-analyses, identify gaps in current knowledge, and highlight potential areas for future studies, thus accelerating the pace of medical research.

Benefits of using specialised language models

Specialised language models offer distinct advantages over general LLMs by focusing explicitly on domain-specific knowledge, terminology, and context. First, SLMs provide higher accuracy and relevance within their targeted fields. Since they are trained on specialised datasets, they understand nuanced terminologies and industry-specific jargon far better than general-purpose LLMs, significantly reducing the likelihood of misinterpretation or irrelevant responses.

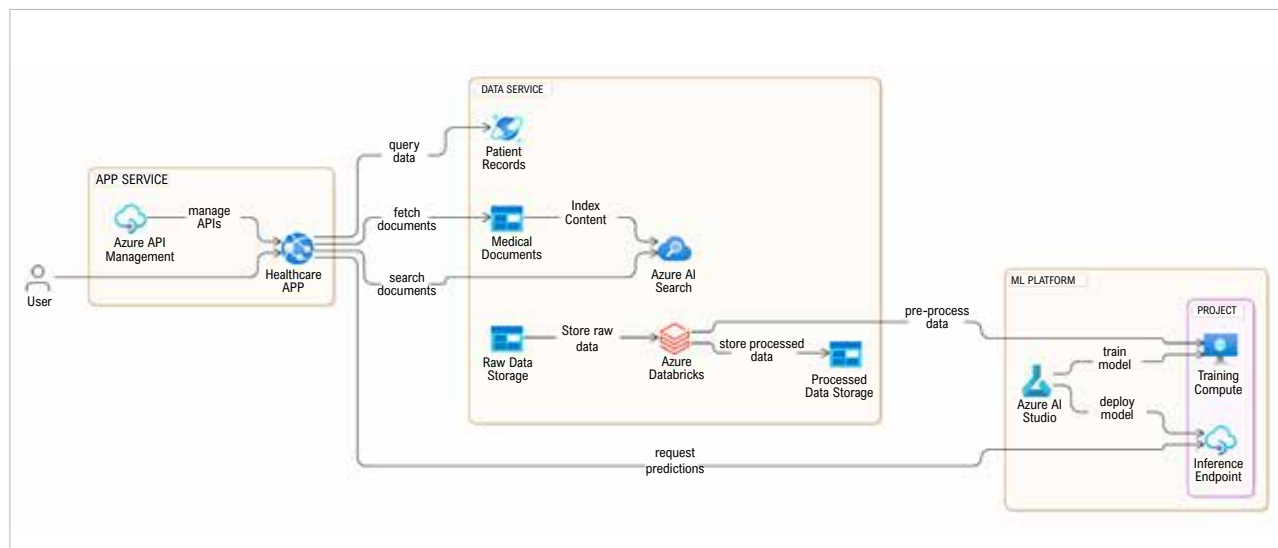


Figure 1: Azure architecture diagram using SLM for healthcare record management